WFLOW-02: Triggers & Event Types

> **Series:** WFLOW — Workflows and Alert Notifications | **Notebook:** 2 of 10 | **Created:** January 2026 | **Last Updated:** 05/21/2026

Event-Driven Workflow Triggers

Triggers determine when workflows execute. This notebook covers all trigger types, detected problem events, metric events, schedules, and custom event triggers.

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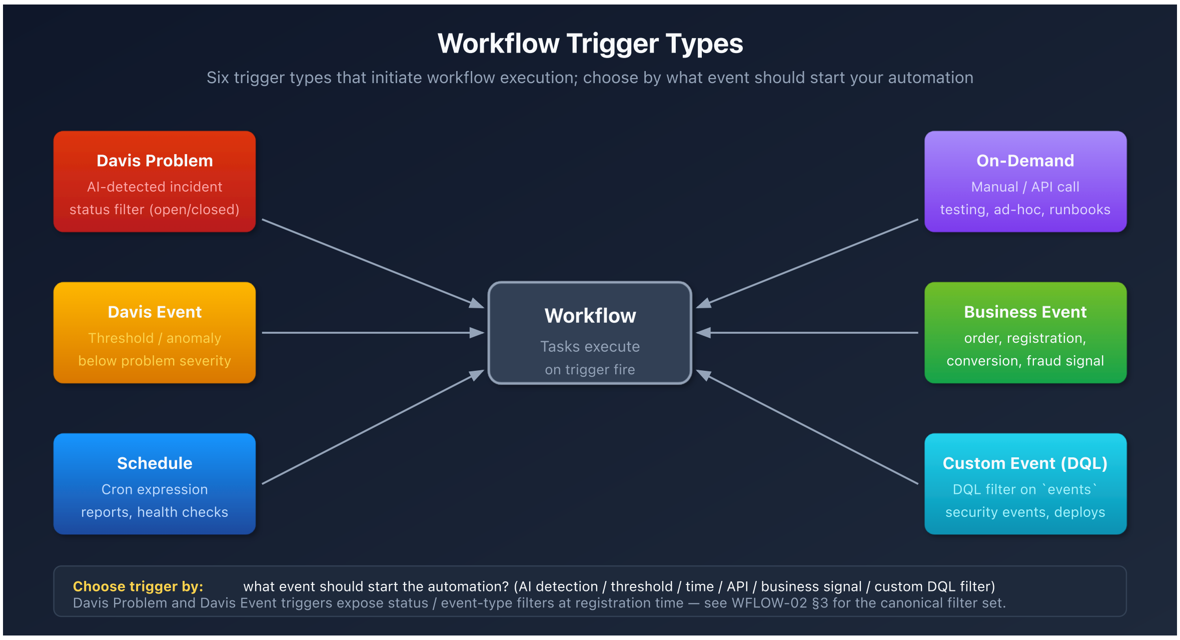
Prerequisites

Requirement	Details
Dynatrace Environment	SaaS with Platform subscription
Permissions	`automation:workflows:write`
Prior Knowledge	**WFLOW-01: Workflow Fundamentals**

1. Trigger Types Overview

Trigger Type	Fires When	Primary Use Case
Detected Problem	Dynatrace Intelligence detects/updates/closes a problem	Alert notifications, incident management
Detected Event	Metric threshold breached	Capacity alerts, proactive notifications
Schedule	Cron expression matches	Reports, health checks, cleanup jobs
On-Demand	Manual execution or API call	Testing, ad-hoc automation
Event	Business/custom event ingested	Business process automation



Choosing the Right Trigger

Scenario	Recommended Trigger
"Notify when a service is slow"	Detected Problem
"Alert when CPU > 90% for 5 mins"	Detected Event
"Send weekly status report"	Schedule

Scenario	Recommended Trigger
"Process order completion events"	Event (bizevents)
"Test my workflow"	On-Demand

2. Detected Problem Trigger

The most common trigger for alert notifications. Fires when Dynatrace Intelligence detects problems.

Trigger Configuration

Setting	Description	Example
Categories	Problem types to include	Infrastructure, Application
Entity Type	Entity types to filter	Service, Host, Process
Management Zones	Scope to specific zones	Production, Checkout
Tags	Entity tags to match	`env:prod`, `team:checkout`

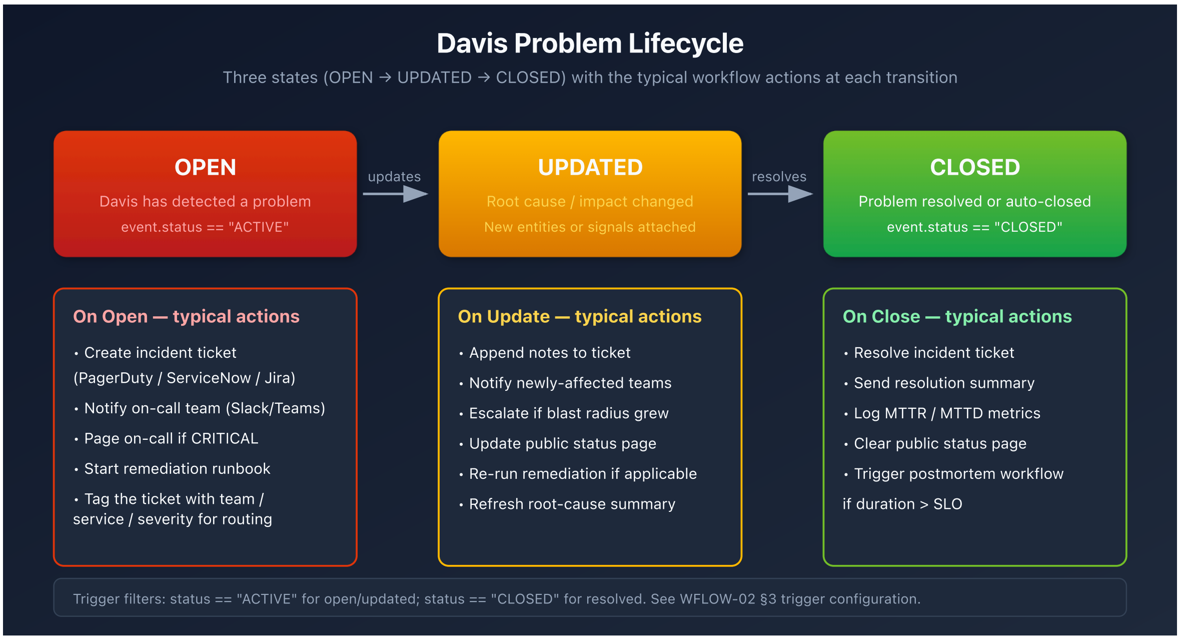
Problem Event Data

When a detected problem triggers, you get access to:

```
{
  "display_id": "P-12345",
  "title": "High response time on checkout service",
  "severity": "CRITICAL",
  "status": "OPEN",
  "start_time": "2026-01-27T10:00:00Z",
  "affected_entity_ids": ["SERVICE-ABC123"],
  "root_cause_entity_id": "HOST-XYZ789",
  "management_zones": ["Production"],
  "impacted_entities": [...],
  "problem_url": "https://env.dynatrace.com/ui/problems/P-12345"
}
```

Problem Lifecycle Events

Event	When	Typical Action
Problem opened	New problem detected	Create incident, notify team
Problem updated	Root cause/impact changed	Update incident notes
Problem closed	Problem resolved	Close incident, send summary



Example: Filter Critical Production Problems

```yaml
trigger:
  type: davis-problem
  config:
    categories:
      - AVAILABILITY
      - PERFORMANCE
    entityTagsMatch: all
    entityTags:
      - key: env
        value: prod
```

3. Detected Event Trigger (Metrics)

Trigger based on metric thresholds without creating a detected problem.

When to Use

- **Proactive alerts** before Dynatrace Intelligence detects a problem
- **Capacity warnings** (disk 80%, memory usage)
- **Business KPIs** (orders/minute, cart abandonment)
- **Custom thresholds** different from baselines

Configuration

Setting	Description
DQL Query	Metric query defining the threshold
Evaluation Frequency	How often to check (1m - 1h)
Alert Condition	When the threshold is breached

Example: CPU Warning at 80%

```
trigger:
  type: davis-event
  config:
    query: |
      timeseries avg_cpu = avg(builtin:host.cpu.usage), by:{dt.entity.host}
      | filter avg_cpu &gt; 80
    evaluationFrequency: "5m"
```

Event Data

```
{
  "metric_key": "builtin:host.cpu.usage",
  "value": 85.5,
  "threshold": 80,
  "entity_id": "HOST-ABC123",
  "triggered_at": "2026-01-27T10:00:00Z"
}
```

4. Schedule Trigger

Execute workflows on a recurring schedule using cron expressions.

Cron Expression Format

```
┌───────────── minute (0-59)
│ ┌───────────── hour (0-23)
│ │ ┌───────────── day of month (1-31)
│ │ │ ┌───────────── month (1-12)
│ │ │ │ ┌───────────── day of week (0-6, Sun=0)
│ │ │ │ │
* * * * *
```

Common Schedules

Schedule	Cron Expression	Description
Every hour	`0 * * * *`	Top of every hour
Daily at 9 AM	`0 9 * * *`	Every day at 9:00
Weekdays at 8 AM	`0 8 * * 1-5`	Mon-Fri at 8:00
First of month	`0 0 1 * *`	1st day, midnight
Every 15 minutes	`*/15 * * * *`	0, 15, 30, 45 past

Example: Daily Health Check Report

```yaml
trigger:
  type: schedule
  config:
    cron: "0 9 * * 1-5"  # Weekdays at 9 AM
    timezone: "America/New_York"
```

Schedule Best Practices

- **Avoid minute 0** - Many workflows run at :00, spread load
- **Use timezones** - Be explicit about timezone

- **Consider rate limits** - Don't schedule too frequently

5. On-Demand Trigger

Manual execution for testing, ad-hoc runs, or API-triggered automation.

Manual Execution

1. Open workflow in editor
2. Click **Run** button
3. Optionally provide input parameters
4. View execution results

API Execution

Trigger workflows programmatically via API:

```
curl -X POST
"https://&lt;env&gt;/platform/automation/v1/workflows/&lt;id&gt;/run" \
  -H "Authorization: Api-Token &lt;token&gt;" \
  -H "Content-Type: application/json" \
  -d '{"params": {"custom_param": "value"}}'
```

Input Parameters

Define custom parameters for on-demand workflows:

```
trigger:
  type: on-demand
  config:
    parameters:
      - name: environment
        type: string
        required: true
      - name: notify_slack
        type: boolean
        default: true
```

Access in tasks:

```
{{ trigger().params.environment }}
{{ trigger().params.notify_slack }}
```

6. Event Trigger (Custom/Business Events)

Trigger on business events or custom events ingested into Grail.

Use Cases

- **Order completed** → Update inventory system
- **User registered** → Send welcome notification
- **Deployment finished** → Run validation tests
- **Custom alert** → External system integration

Configuration

```
trigger:
  type: event
  config:
    eventType: "com.company.order-completed"
    filterQuery: |
      event.type == "com.company.order-completed"
      AND order.total &gt; 1000
```

Sending Business Events

Ingest events via API:

```
curl -X POST "https://&lt;env&gt;/api/v2/bizevents/ingest" \
  -H "Authorization: Api-Token &lt;token&gt;" \
  -H "Content-Type: application/json" \
  -d '{
    "type": "com.company.order-completed",
    "data": {
      "order_id": "ORD-12345",
      "customer": "ACME Corp",
      "total": 1500.00
    }
  }'
```

Event Data

Access event fields in tasks:

```
{{ event()["data"]["order_id"] }}
{{ event()["data"]["customer"] }}
{{ event()["data"]["total"] }}
```

7. Trigger Data and Expressions

Accessing Trigger Data

Expression	Returns	Example
`{{ event() }}`	Full event object	`{"title": "...", ...}`
`{{ event()["field"] }}`	Specific field	`"High response time"`
`{{ trigger() }}`	Trigger metadata	`{"type": "davis-problem"}`
`{{ trigger().params }}`	On-demand params	`{"env": "prod"}`

Common Event Fields by Trigger Type

Detected Problem:

```
{{ event()["display_id"] }}           # P-12345
{{ event()["title"] }}                # Problem title
{{ event()["severity"] }}             # CRITICAL, HIGH, MEDIUM, LOW
{{ event()["status"] }}               # OPEN, RESOLVED
{{ event()["affected_entity_ids"] }}  # Array of entity IDs
{{ event()["root_cause_entity_id"] }} # Root cause entity
{{ event()["management_zones"] }}     # Array of MZ names
{{ event()["problem_url"] }}          # Link to problem
```

Detected Event (Metric):

```
{{ event()["metric_key"] }}    # Metric identifier
{{ event()["value"] }}         # Current value
{{ event()["threshold"] }}     # Threshold that was breached
{{ event()["entity_id"] }}     # Affected entity
```

Schedule:

```
{{ trigger()["scheduled_time"] }}  # When scheduled to run
{{ trigger()["actual_time"] }}     # When actually started
```

8. Davis Problem Event Payload Reference

> **Verified against Dynatrace Workflows trigger reference (DT docs), May 2026.** Payload field names and lifecycle semantics can drift sprint-to-sprint — re-verify against your tenant version before building load-bearing routing logic on uncommon fields.

Workflows on a Detected Problem trigger receive a single `event` object representing the Davis problem at the moment the trigger fired. Sections 2 and 7 above show common access patterns; this section is the full reference: every field commonly seen in the payload, when it appears, how it joins to DQL, and which downstream notebooks consume it.

8.1. Top-Level Field Reference

Field	Type	Always present?	Notes
`event.kind`	string	Yes	Always `"DAVIS_PROBLEM"` for this trigger. Distinguishes from `DAVIS_EVENT` (raw signals).
`display_id`	string	Yes	Human-facing problem ID (`"P-12345"`). Use in notifications and ticket subjects.
`problem_id`	string	Yes	Internal problem ID. Use to **join to `fetch dt.davis.problems`** (see §8.3).

Field	Type	Always present?	Notes
`event.id`	string	Yes	Event-record ID for this specific lifecycle update (different per OPEN/UPDATE/CLOSE firing).
`title` / `event.name`	string	Yes	Short problem title — `"High response time on checkout service"`. Both names appear; `event.name` is the DQL-canonical form.
`severity` / `event.category`	string	Yes	One of `AVAILABILITY`, `ERROR`, `SLOWDOWN`, `RESOURCE`, `CUSTOM_ALERT`, `MONITORING_UNAVAILABLE`, `INFO`. Older fields may also surface `CRITICAL` / `HIGH` / `MEDIUM` / `LOW` semantic levels.
`status`	string	Yes	`ACTIVE` (open) or `CLOSED`. Older notebooks and docs sometimes show `OPEN` / `RESOLVED` — both names occur depending on surface.
`event.status_transition`	string	On UPDATE/CLOSE	`CREATED`, `UPDATED`, or `CLOSED` — tells the workflow *which lifecycle event* triggered it.
`start_time`	timestamp (ISO 8601)	Yes	When the problem opened.
`end_time`	timestamp (ISO 8601)	On CLOSE only	When the problem closed. Null/missing while active.

Field	Type	Always present?	Notes
`affected_entity_ids`	string[]	Yes	All entities Davis correlated to this problem (services, hosts, processes, etc.).
`root_cause_entity_id`	string	When determined	Single entity Davis identified as root cause. May be empty/null on initial OPEN before causation analysis completes.
`impacted_entities`	object[]	Yes	Richer per-entity records — typically `{ id, name, type }`. Use when you need the entity *name* without an extra lookup.
`affected_entity_types`	string[]	Yes	Distinct entity types touched by this problem (`["SERVICE", "HOST"]`). Useful for routing without enumerating entity IDs.
`management_zones`	string[]	Yes	Management-zone names this problem touches. Drives team routing in WFLOW-04.
`entity_tags`	object[]	When tagged	Tags on the affected entities (`[{"key": "team", "value": "checkout"}, ...]`). Custom-tag routing reads from here.
`problem_url`	string	Yes	Deep-link to the Davis Problems-app page for this problem. Always include in notifications.
`event.description`	string	Variable	Longer human-readable description; not always populated. Treat as optional.

> **Sources:** Workflow triggers (DT docs), Workflow reference / Jinja expressions (DT docs), Davis Problems app (DT docs). **Derived:** the OPEN vs UPDATE vs CLOSE field-presence column synthesizes the trigger reference with observed payloads — `end_time` and `event.status_transition` are the load-bearing differentiators across lifecycle stages.

8.2. Lifecycle Field Presence

A single workflow can fire on problem **CREATED**, **UPDATED**, and **CLOSED**. The payload shape differs at each stage — guard logic that assumes all fields are populated will silently mis-route close events.

Field	CREATED	UPDATED	CLOSED
`display_id`, `problem_id`, `title`	✓	✓	✓
`severity`, `status`	✓	✓	✓ (`CLOSED`)
`event.status_transition`	`CREATED`	`UPDATED`	`CLOSED`
`start_time`	✓	✓	✓
`end_time`	—	—	✓
`root_cause_entity_id`	sometimes empty	usually populated	populated
`affected_entity_ids`	initial set	may expand	final set
`entity_tags`	from initial entities	may expand	final set
`problem_url`	✓	✓	✓

In community practice the safest routing pattern is: branch on `event.status_transition` first (and fall back to `status` if absent), then read only the fields guaranteed for that branch.

8.3. Joining the Payload to DQL

The workflow payload is a snapshot. For longer-window analysis (problem history, MTTR, fleet-wide patterns), join the payload's `problem_id` to the canonical Grail table:

```
// Hydrate the workflow payload against the full Davis problems record.
// Substitute {{ event()["problem_id"] }} via a DQL workflow task.
fetch dt.davis.problems, from:-7d
| filter event.id == "{{ event()['problem_id'] }}"
| fields display_id,
         event.name,
         event.category,
         event.status,
         event.start,
         event.end,
         affected_entity_ids,
         root_cause_entity_id,
         management_zones
| limit 1
```

Why `dt.davis.problems` and not `dt.davis.events`: `dt.davis.problems` is the canonical problems data object; `dt.davis.events` carries only `DAVIS_EVENT` records (raw signals that *feed* problem detection, not the problems themselves). Querying `dt.davis.events` for problem records returns zero rows on modern tenants.

For aggregate views (e.g., "all problems for the same entity this week") swap the filter:

```
fetch dt.davis.problems, from:-7d
| filter in("{{ event()['affected_entity_ids'][0] }}", affected_entity_ids)
| summarize problem_count = count(), by:{event.category, event.status}
| sort problem_count desc
```

8.4. Common Access Patterns

```
# Display ID and link — every notification
{{ event()["display_id"] }}
{{ event()["problem_url"] }}

# Severity-based routing (see WFLOW-04 §3 routing-by-severity)
{% if event()["severity"] == "AVAILABILITY" %} page on-call
{% elif event()["severity"] == "ERROR" %}        notify channel
{% else %}                                         log only
{% endif %}

# Lifecycle branch — distinguishes open/update/close in the same workflow
{% if event()["event.status_transition"] == "CLOSED" %} close ticket
{% else %}                                                 update ticket
{% endif %}

# Custom-tag access (entity_tags is an array of {key, value} objects)
{% for tag in event()["entity_tags"] %}
  {% if tag["key"] == "team" %}{{ tag["value"] }}{% endif %}
{% endfor %}

# Entity-type filtering — route by what kind of entity is involved
{% if "SERVICE" in event()["affected_entity_types"] %} service-team channel
{% elif "HOST" in event()["affected_entity_types"] %}   infra channel
{% endif %}

# Root cause name (no extra lookup needed — impacted_entities carries names)
{% for ent in event()["impacted_entities"] %}
  {% if ent["id"] == event()["root_cause_entity_id"] %}{{ ent["name"] }}{%
endif %}
{% endfor %}
```

8.5. Cross-References

- **WFLOW-04 §3 *Routing by Severity*** consumes `severity` and `management_zones` from this payload to fan notifications out to team channels.
- **WFLOW-04 §4 *Routing by Team/Service*** reads `entity_tags` (custom tags like `team:checkout`) and `affected_entity_types`.
- **WFLOW-05 *Incident Management*** uses `display_id`, `problem_url`, `status`, and `start_time` to create and update tickets through the lifecycle.
- **WFLOW-07 *Remediation*** branches on `root_cause_entity_id` and `title` to pick the right runbook task, and joins `problem_id` to `dt.davis.problems` for richer

context before acting.

8.6. Honest Caveats

- **Field-name drift:** Some fields surface under two names (`title` ↔ `event.name`, `severity` ↔ `event.category`, `status` value `OPEN` vs `ACTIVE`). Older workflow templates and docs frequently use the older form; new workflows should prefer the `event.*` canonical names because they match the DQL field surface in `dt.davis.problems`.
- **`status` value vocabulary:** Modern Davis records use `ACTIVE` / `CLOSED`; some legacy notebooks (including earlier sections of this notebook) still show `OPEN` / `RESOLVED`. Both can appear in payloads depending on tenant version — guard with a set check rather than `==`.
- **Re-verify before load-bearing logic:** This reference is dated 05/21/2026. The Workflows trigger schema is product surface and can change in a sprint. Spot-check a real payload (use a logging task or send-to-Slack-as-debug task) in your tenant before relying on field availability in production routing.

Query Detected Problems

```
// Recent detected problems that could trigger workflows
fetch events, from: now() - 24h
| filter event.kind == "DAVIS_PROBLEM"
| fields timestamp,
         display_id,
         event.name,
         severity,
         status,
         affected_entity_ids,
         root_cause_entity_id
| sort timestamp desc
| limit 20
```

```
// Problem count by severity (last 7 days)
fetch events, from: now() - 7d
| filter event.kind == "DAVIS_PROBLEM"
| filter status == "OPEN"
| summarize problem_count = count(), by:{severity}
| sort problem_count desc
```

```
// Business events that could trigger workflows
fetch bizevents, from: now() - 24h
| summarize event_count = count(), by:{event.type}
| sort event_count desc
| limit 20
```

Next Steps

Now that you understand triggers, learn to send notifications:

Recommended Path

1. **WFLOW-03: Alert Notification Basics** - Slack, Teams, email notifications
2. **WFLOW-04: Advanced Notification Routing** - Conditional routing
3. **WFLOW-07: Problem-Triggered Remediation** - Auto-remediation patterns

Key Takeaways

- **Detected Problem** triggers for AI-detected incidents
- **Detected Event** triggers for metric thresholds
- **Schedule** triggers for recurring tasks
- **On-Demand** triggers for testing and API integration
- **Event** triggers for business events
- Use expressions like `{{ event()["field"] }}` to access data

Summary

In this notebook, you learned:

- All five trigger types and when to use each
- How to configure Detected Problem triggers with filters
- How to set metric thresholds with Detected Event triggers
- Cron expressions for schedule triggers
- On-demand execution via UI and API

- Business event triggers for custom automation
- How to access trigger data in expressions

References

- Workflow triggers (DT docs)
- Workflow reference / Jinja expressions (DT docs)
- Davis Problems app (DT docs)
- Business analytics umbrella (DT docs)
- Workflows umbrella (DT docs)
- Cron expression sandbox (crontab.guru)

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